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Contribution of Direct and Indirect Exposure to Human Serum Concentrations of Perfluorooctanoic Acid in an Occupationally Exposed Group of Ski Waxers

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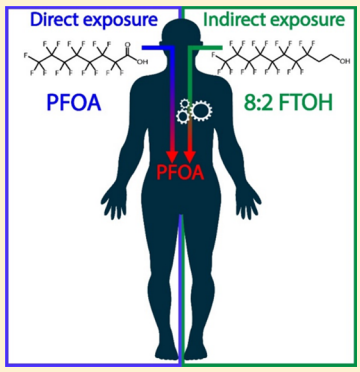
Supporting Information

ABSTRACT: The contribution of direct (i.e., uptake of perfluorooctanoic acid (PFOA) itself) and indirect (i.e., uptake of 8:2 fluorotelomer alcohol (FTOH) and metabolism to PFOA) exposure to PFOA serum concentrations was investigated using a dynamic one-compartment pharmacokinetic (PK) model. The PK model was applied to six occupationally exposed ski waxers for whom direct and indirect exposures via inhalation were characterized using multiple measurements with personal air sampling devices. The model was able to predict the diverging individual temporal trends of PFOA in serum with correlation coefficients of 0.82−0.94. For the four technicians with high initial concentrations of PFOA in serum (250−1050 ng/mL), the ongoing occupational exposure (both direct and indirect) was of minor importance and net depuration of PFOA was observed throughout the ski season. An estimated average intrinsic elimination half-life of 2.4 years (1.8−3.1 years accounting for variation between technicians and model uncertainty) was derived for these technicians. The remaining two technicians, who had much lower initial serum concentrations (10−17 ng/mL), were strongly influenced by exposure during the ski season with indirect exposure contributing to 45% of PFOA serum concentrations. On the basis of these model simulations, an average metabolism yield of 0.003 (molar concentration basis; uncertainty range of 0.0006−0.01) was derived for transformation of 8:2 FTOH to PFOA. An uncertainty analysis was performed, and it was determined that the input parameters quantifying the intake of PFOA were mainly responsible for the uncertainty of the metabolism yield and the initial concentration of PFOA in serum was mainly contributing to the uncertainty of estimated serum half-lives.



1. INTRODUCTION

Perfluoroalkyl carboxylic acids (PFCAs) are fluorinated surfactants that have been produced and used in various industrial applications since the 1950s.[1] Among PFCAs, perfluorooctanoic acid (PFOA) is of particular concern for the environment and for human health as a result of its global distribution,[2] persistence, bioaccumulation potential,[3,4] and toxicity in animal models.[5] Increased serum cholesterol,[6] decreased fetal growth,[7] and reduced humoral immune response to routine childhood immunizations[8] have been positively associated with PFOA serum levels in the background exposed population. For occupationally exposed workers and populations with elevated exposures from consumption of contaminated drinking water, associations between exposure to PFOA and several diseases have been observed, namely, increased incidence of kidney and testicular cancer,[9,11] ulcerative colitis,[10,11] clinically defined high cholesterol,[11] thyroid disease,[11] and pregnancy-induced hypertension.[11] The pharmacokinetics of PFOA in organisms are distinct from the majority of persistent organic contaminants. Unlike neutral hydrophobic contaminants that preferably partition to adipose tissue, the high affinity for serum albumin[9] and liver-fatty acid binding proteins[10] together with renal tubular reabsorption processes[11] have been identified as the main mechanisms of PFOA accumulation in humans. As a consequence of its long-term release in large amounts and its long elimination half-life in humans, PFOA has been detected in serum samples of occupationally exposed humans[12] and in the general population worldwide.[13−16]

Contemporary background human exposure to PFOA occurs through multiple pathways with contaminated food,[17] drinking water,[18] and ingestion of indoor dust[19] being the most important exposure pathways for adults. In addition to direct exposure (uptake of PFOA itself), it has been hypothesized that the body burden of PFOA is influenced by the exposure to precursor compounds which are metabolized in the body to PFOA after uptake (hereafter referred to as indirect exposure).[17,20−22] Among these precursors, 8:2 fluorotelomer alcohol (8:2 FTOH) has been extensively studied since it has been one of the major building blocks of fluorotelomer-based consumer products. Although several studies showed that 8:2

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FTOH is metabolized into PFOA in rats,[23−25] mice and trout,[26] the quantification of the metabolism yield in humans remains challenging given the large interspecies variability in metabolism yields and difficulties to extrapolate yields from *in vitro* experiments on human hepatocytes.[26] Furthermore, most of the background exposure studies that assessed the uptake pathways and kinetics of PFOA in the general human population considered direct exposure,[14,19,27] but rarely considered indirect exposure sources. Reasons for difficulties in quantifying the contribution of indirect exposure are (1) the lack of exposure data for the various precursors and (2) multiple exposure pathways all making important contributions and being rarely studied simultaneously.

The most convincing evidence to date that exposure to FTOHs yields PFCAs in human serum can be found from a unique data set on occupationally exposed professional ski waxers. In the work by Nilsson et al.,[28,29] extremely high exposures to 8:2 FTOH via inhalation of indoor air were linked to elevated concentrations of PFOA in serum samples of the ski technicians. Follow-up studies also confirmed the presence of numerous FTOH metabolism intermediates in the serum samples of ski wax technicians,[29] which have been observed in controlled biotransformation experiments.[23,30] Nevertheless, no attempts were made to quantitatively reconstruct the measured levels of PFOA in serum from the wealth of external exposure data (of both PFOA and 8:2 FTOH in air) or estimate the metabolism yield of 8:2 FTOH.

In this study, we aimed to assess the contributions of direct and indirect exposure to the serum concentrations of PFOA using the occupational exposure data from Nilsson et al.[28,29] To accomplish this objective, a dynamic one-compartment pharmacokinetic (PK) model was employed to reconstruct time trends of PFOA in serum from measurements of PFOA and 8:2 FTOH concentrations in air and dust reported by Nilsson et al.[28,29] By fitting the model to the biomonitoring data, the metabolism yield (Y_{meta}), which reflects the amount of PFOA produced from a given amount of 8:2 FTOH entering the systemic circulation, and the elimination half-life of PFOA were estimated. This study is the first to perform individual-based model predictions of PFOA exposure and quantify the metabolism yield of 8:2 FTOH conversion to PFOA in humans considering the entire physiological system (i.e., *in vivo*).

2. METHOD

2.1. Model Structure and Assumptions. PK modeling is a well-established approach to study the uptake, distribution, and elimination of therapeutic drugs that can also be applied to probe the intake and elimination of environmental contaminants. Several PK models of various complexity have been developed and applied to predict the serum and tissue concentrations of PFOA in humans.[27,31] In this study, the serum concentrations of PFOA were simulated using a dynamic one-compartment model, where serum concentrations were calculated as a function of total intake, elimination rate, and volume of distribution (Figure 1). The high temporal resolution of data provided by Nilsson et al.[28,29] allowed dynamic modeling simulations to be undertaken. Even though the use of a one-compartment PK model for PFOA has been previously criticized,[27] increasing the complexity of the model was not necessary for the following reasons: (1) the serum concentrations of PFOA in ski waxers were too low to trigger saturation processes[32] and (2) including a higher amount of input parameters to describe tissue distribution was not deemed necessary for this study as only blood samples were available for model evaluation.[28,29]

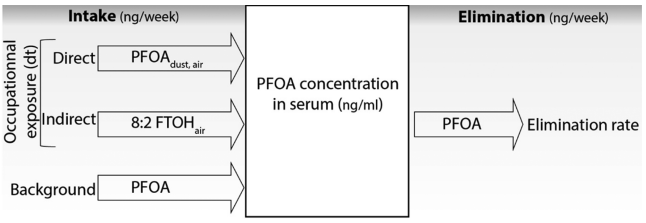

Figure 1. Dynamic one-compartment mass balance model with intake from different exposure pathways (direct, indirect, and background) and the elimination rate of PFOA.

Occupational exposure of ski wax technicians was assumed to be primarily via inhalation of air and ingestion of suspended particles since the heating and application of waxes and powders on ski soles produces a large amount of particles and fumes containing high levels of PFOA and 8:2 FTOH.[28] The hand-to-mouth exposure pathways was not included due to lack of information on the amount of contaminated dust or product transferred onto the hands during waxing activities. Background exposure was also included in the total intake of chemical. Although the background exposure was orders of magnitude lower than the occupational exposure of the ski wax technicians, the inclusion of this process may be important to correctly describe temporal trends during the off-season.[33] Uptake of PFOA and 8:2 FTOH from the contact of the wax with the epidermis was neglected as dermal absorption has been shown to be minor.[24,34]

The model was parametrized using occupational exposure data with repeated measurements from ski wax technicians that were highly exposed to PFOA and 8:2 FTOH.[28,29] Due to differences in employment length and exposure history, the measured initial serum concentrations of PFOA varied from 10 to 1050 ng/mL and different temporal PFOA serum trends over time were observed among the technicians. Given this large variability, the model was applied to each technician individually. In order to investigate the contribution of indirect exposure from the direct exposure, two model scenarios were applied: (1) the technician was exposed to PFOA only (i.e., measured direct intake of PFOA from ski-waxing and estimated background intake of PFOA); (2) the technician was exposed to PFOA and 8:2 FTOH from ski-waxing and to the estimated background intake of PFOA.

2.2. Model Equations. The concentration of PFOA in serum (C_{serum} in ng/mL) is a function of the direct and indirect occupational intake of PFOA, background intake of PFOA, elimination rate of PFOA in serum, and the apparent volume of distribution, as presented in Figure 1. Following this model structure, a first-order differential equation was used to describe the mass balance of an individual exposed to both PFOA and 8:2 FTOH (eq 1).

$$\frac{dC_{serum}}{dt} = \frac{I_{direct}(t) + I_{indirect}(t) + I_{bkg}}{V_d \times B_w} - C_{serum}(t) \times k_e \quad (1)$$

where I_{direct} is the direct intake of PFOA, $I_{indirect}$ is the indirect intake through the uptake and metabolism of 8:2 FTOH, and I_{bkg} is the background intake of PFOA (all intakes in ng PFOA/week). Both I_{direct} and $I_{indirect}$ varied over time according to measured levels of PFOA and 8:2 FTOH in external exposure

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media, while I_{bkg} was assumed to be constant. The volume of distribution (V_d in mL/kg) is a proportionality parameter between the total amount of PFOA in the body and its concentration in serum. Because the volume of distribution was expressed here in mL/kg, V_d was multiplied by the body weight (B_w in kg) of each participant. The elimination of PFOA from serum was represented by the elimination rate (k_e in 1/week). The model was based on a discrete-time approach, and the time step dt was set as 1 week.

I_{direct}, $I_{indirect}$, I_{bkg}, and k_e as presented in eq 1 were calculated following eqs 2, 3, 4, and 5.

$$I_{direct}(t) = (C_{dust}(t) \times C_{d_{PFOA}}(t) \times K_i \times E_{gi} + C_{airPFOA}(t) \times K_i \times E_i) \times W_h \quad (2)$$

$$I_{indirect}(t) = C_{airFTOH}(t) \times K_i \times E_i \times W_h \times Y_{meta} \quad (3)$$

$$I_{bkg} = D_{bkg} \times E_{gi} \times B_W \times t \quad (4)$$

$$k_e = \frac{0.693}{t_{1/2}} \quad (5)$$

For the calculation of I_{direct} (eq 2), both PFOA present as vapor in air ($C_{airPFOA}$, in ng/m^3) and sorbed to air particles were considered. The latter was calculated from the measured concentration of PFOA sorbed to suspended dust (C_{dPFOA}, in ng/mg dust) and the concentration of suspended dust particles measured in air (C_{dust}, in mg/m^3). The gaseous PFOA was assumed to be taken up via the lungs, and the amount of chemical absorbed depended on the volume of respired air, K_i (m^3/hour), and the absorption efficiency of the chemical through the respiratory epithelium (E_i, unitless). The particle fraction was assumed to be ingested (i.e., inhaled and coughed up to the trachea followed by ingestion with efficient absorption in the gut), and the amount of PFOA from dust particles absorbed was determined by K_i and by its absorption efficiency through the intestinal epithelium (E_{gi}, unitless). Due to its high vapor pressure, 8:2 FTOH is primarily present in the gas phase.[28,29,35] Subsequently, $I_{indirect}$ (eq 3) included only the gas phase air concentration of 8:2 FTOH ($C_{airFTOH}$, in ng/m^3). The total intake of PFOA from indirect and direct exposure was multiplied by the amount of working hours per week (W_h, in hours/week) to give the weekly intake of chemical from occupational exposure. Since the metabolism rate of 8:2 FTOH is dramatically faster than the total elimination rate of PFOA,[25,26] instantaneous metabolism to PFOA was assumed. The metabolism yield (Y_{meta}, unitless) represents the fraction between the amount of formed PFOA from metabolism of 8:2 FTOH and the amount of absorbed 8:2 FTOH. I_{bkg} was calculated (eq 4) from the average daily intake of PFOA in the general population (D_{bkg}, in ng/kg/d) and was normalized to obtain the amount of PFOA assimilated per individual over a week (time t = 7 d/week). The elimination rate of PFOA (k_e, in/week) was calculated from the elimination half-life ($t_{1/2}$, in week) of PFOA (eq 5). Except for Y_{meta} and $t_{1/2}$, which were calibrated in this study, all input parameters were obtained from the individual specific data collected by Nilsson et al.'s study[28,29] or other literature (described below).

2.3. Selection of Input Parameters. *2.3.1. External Exposure.* Concentrations of 8:2 FTOH and PFOA in air and dust were taken directly from Nilsson et al.'s study.[28,29] Generally, the working season of the ski waxers started at the end of November/beginning of December and continued until the end of April/beginning of May followed by 6 months off-duty until the next ski season. As summarized in Figure A2-1 in the Supporting Information, the study of Nilsson et al.[28,29] followed this seasonality over 2 years, from September 2007 until April 2009. During the off-duty periods, it was assumed that the technicians were not in contact with ski waxes and were subject to background exposure only. During working seasons, direct and indirect exposure was assumed to take place together with background exposure. Concentrations of PFOA and 8:2 FTOH in air and suspended particles were monitored during the two working seasons, from December 2007 to April 2008 and from November 2008 to April 2009. With the exception of February 2009, when no measurements occurred, samples of air and suspended dust were collected once a month during both exposure periods, whenever cross-country ski competitions were taking place. The selection and treatment of monitoring data used to estimate direct and indirect intakes is described in more detail in Appendix 3 of the Supporting Information.

The inhalation rate (K_i in m^3/hour), which is proportional to the physical activity of the person, was obtained from the literature. It was assumed that the application of wax on skis required "movements of light intensity", according to US EPA classification of activity levels.[36] The corresponding K_i value of 0.72 m^3/hour was applied in the model. Due to a lack of quantitative uptake studies on inhalation, the pulmonary absorption (E_i, unitless) of PFOA and 8:2 FTOH was assumed to be 100%, as previously done in other studies.[25,37,38] Following the high gastro-intestinal absorption fraction of PFOA in animals,[39,40] the intestinal absorption (E_{gi}, unitless) was set as 0.9. The body weight of individual technicians is given in Appendix 2 of the Supporting Information, and all technicians were assumed to work 30 h a week in the ski wax cabin.

Because food and beverage are believed to be the major source of PFOA for the general population,[19,41−44] the background intake of PFOA used in this study considered exposure from diet and drinks only. The minimum and maximum daily intakes of PFOA in Nordic countries have been estimated to be between 0.2 and 0.7 ng/kg/day.[35,44,45] Consequently, an intake of 0.350 ng/kg/day estimated by Vestergren et al.[45] was used for the calculation of I_{bkg}.

2.3.2. Volume of Distribution. For a one-compartment model, V_d (in mL/kg) describes the equivalent volume of serum to which the chemical partitions. In this work, V_d was obtained from literature values. Even though interspecies variability has been observed, V_d estimated from animal dosing studies[46−50] has been successfully applied to human one-compartment PK modeling.[51,52] A V_d of 200 mL/kg was used in this study. This value is close to the averaged V_d taken from various studies on animals and humans (i.e., 194 mL/kg).[46−50,53,54] As mentioned in Section 2.2, distribution parameters for 8:2 FTOH in blood were not necessary since metabolism was assumed to happen prior to the distribution of 8:2 FTOH in the blood compartment.

2.3.3. Calibration of Y_{meta} and $t_{1/2}$. Blood concentrations (C_{serum}) used for the calibration of Y_{meta} and $t_{1/2}$ were obtained from Nilsson et al.[28,29] Measurements were taken once a month during the working and off-duty seasons over the two-year study, provided that the technicians were available for sampling. Since the biomonitoring data were based on whole blood samples, they were adjusted to serum concentrations according to the method of Ehresman et al.[55] The latter observed that

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serum concentrations of PFOA are ca. 2 times higher than whole blood concentrations.

The combination of external exposure and biomonitoring data from ski wax technicians was used to estimate the metabolism yield of 8:2 FTOH to PFOA and the elimination half-life of PFOA. A sufficient amount of data covering the individual exposure from September 2007 to April 2009 was necessary to allow dynamic modeling of each technician and a proper calibration of Y_{meta} and $t_{1/2}$. As a consequence, even though a total of 11 technicians were monitored by Nilsson et al.,[28,29] only six could be considered in this study. More detailed information on each of the six technicians is available in Table A2-1 of the Supporting Information (the initial numbering of technicians in Nilsson et al.[28,29] was kept). According to their levels in serum, the six technicians were classified as follows: (1) technicians (1−2) with low initial concentration of PFOA in serum (initial concentration <20 ng/mL); (2) technicians (3, 4, 6, 8) with high initial concentration of PFOA in serum (initial concentration >200 ng/mL). For all technicians, one-time serum measurements were taken in 2010 and 2011, at the end of the working season. Consequently, the model predicted serum concentrations based on a four-year period. At $t = 0$, C_{serum} in eq 1 was set as the initial serum concentration measured in each technician at the beginning of the study. Except for technician 6, for whom samples were available from November 2007 only, this value corresponded to the biomonitoring data in September 2007, prior to the winter season 2007−2008.

Y_{meta} was fitted on the biomonitoring data of the technicians with low concentration, who were influenced by the ongoing occupational exposure, and $t_{1/2}$ (in units of week) was fitted on the biomonitoring data of the technicians with high concentration, on whom the external exposure had a low impact. The least-square optimization method[56] was used to minimize the sum of squared residual weighted (SSRW) and provided the closest model predictions to the respective experimental data. The best estimated Y_{meta} or $t_{1/2}$ corresponded to the simulation with the closest coefficient of determination (R^2) to 1.

When Y_{meta} was determined as the fitted output parameter, $t_{1/2}$ needed to be set as a fixed input parameter. $t_{1/2}$ has been estimated by various studies on highly exposed workers and communities with elevated exposure from contaminated drinking water.[52,57,58] Following these findings, an estimated elimination half-life of 2.4 years was applied in our model. When $t_{1/2}$ was determined as the fitted output parameter, Y_{meta} was set as the average of the Y_{meta} estimated on the ski wax technicians.

2.4. Evaluative Model Simulations of Long-Term Seasonal Exposure. Two evaluative simulations were performed to illustrate the typical long-term (decadal) trends of PFOA serum concentrations in ski wax technicians during their careers and under different exposure conditions. Specific assumptions, which sometimes deviated from the observations by Nilsson et al.,[28,29] were made to highlight how the internal levels respond to long-term exposure and changes in occupational exposure. The first simulation was performed to illustrate the PFOA serum concentrations of a ski wax technician employed during the winter ski seasons every year between the ages of 35 and 60. The concentrations of PFOA and 8:2 FTOH in air were assumed to be 1×10^4 and 1×10^5 ng/m^3, respectively, and the concentration of PFOA in suspended dust was assumed to be 1×10^2 ng/m^3. These concentrations, that were assumed to be constant during all working seasons, were chosen in an attempt to reproduce the serum concentration of technician 8 (i.e., 1050 ng/mL). Background exposure was assumed to be 171 ng/week throughout the technician's life. After retirement, occupational exposure ceases and only background exposure was assumed to occur.

In the second simulation, two male ski-wax technicians started working at different points in time, under different occupational exposure conditions. As in the first evaluative simulation, the first technician worked seasonally from 1990 until 2025. The second technician started working in the same cabin 15 years later, from 2005 and onward. The seasonal occupational exposure was drastically decreased in 2005 to reflect exposure mitigating strategies such as wearing protective equipment (face masks and gloves) and working in a highly ventilated cabin. The concentration in air for this period was set as 1×10^3 and 1×10^4 ng/m^3 for PFOA and 8:2 FTOH, respectively, and 1×10 ng/m^3 for the concentration of PFOA in suspended dust. For the first 15 years (1990−2005), three different exposure conditions were defined (i.e., 2-, 5-, and 10-fold higher exposure than the levels set after 2005) in order to reproduce similar serum levels as the technicians with high internal concentrations studied by Nilsson et al.[28,29]

2.5. Sensitivity and Uncertainty Analysis. Following the principles of Good Modeling Practice,[59] a sensitivity analysis was performed of the model for each of the six technicians. All model input parameters were decreased by 0.1%, as suggested by MacLeod et al.,[60] to monitor their influence on predicted metabolism yield and elimination half-life. The sensitivity ratio S_I was calculated for each of the input parameters, according to eq 6:

$$S_I = \frac{(y_2 - y_1)/y_1}{(x_2 - x_1)/x_1} \tag{6}$$

S_I provides an insight into the sensitivity of the model output y for a given change in the input parameter x.

The uncertainty in the fitted metabolism yield and elimination half-life was quantified following the first-order error propagation method presented in MacLeod et al.[60] Briefly, a confidence factor (Cf_I), which is a measure of variance (e.g., a confidence factor of 2 means that 95% of the values are found between 1/2 and 2 times the median), was determined for each input parameter used in the model (see section A6-2 in Appendix 6 of the Supporting Information). The uncertainty of the output result is a function of both its sensitivity to a given input parameter and the uncertainty of the input parameter. The contribution of each input parameter j to the variance in the output result was therefore calculated for each technician (eq 7):

$$\frac{S_{I,j}{}^2(\ln Cf_{I,j})^2}{\sum_{j=1}^{n} S_{I,j}{}^2(\ln Cf_{I,j})^2} \tag{7}$$

Finally, the margin of error was investigated for each estimated metabolism yield and elimination half-life. To do so, the confidence factor of the output parameters (Cf_o, corresponding to the Y_{meta} and the $t_{1/2}$) was calculated from the n individual input parameters according to eq 8.

$$Cf_o = \exp[S_{I,1}{}^2(\ln Cf_{I,1})^2 + S_{I,2}{}^2(\ln Cf_{I,2})^2 + \cdots + S_{I,n}{}^2(\ln Cf_{I,n})^2]^{1/2} \tag{8}$$

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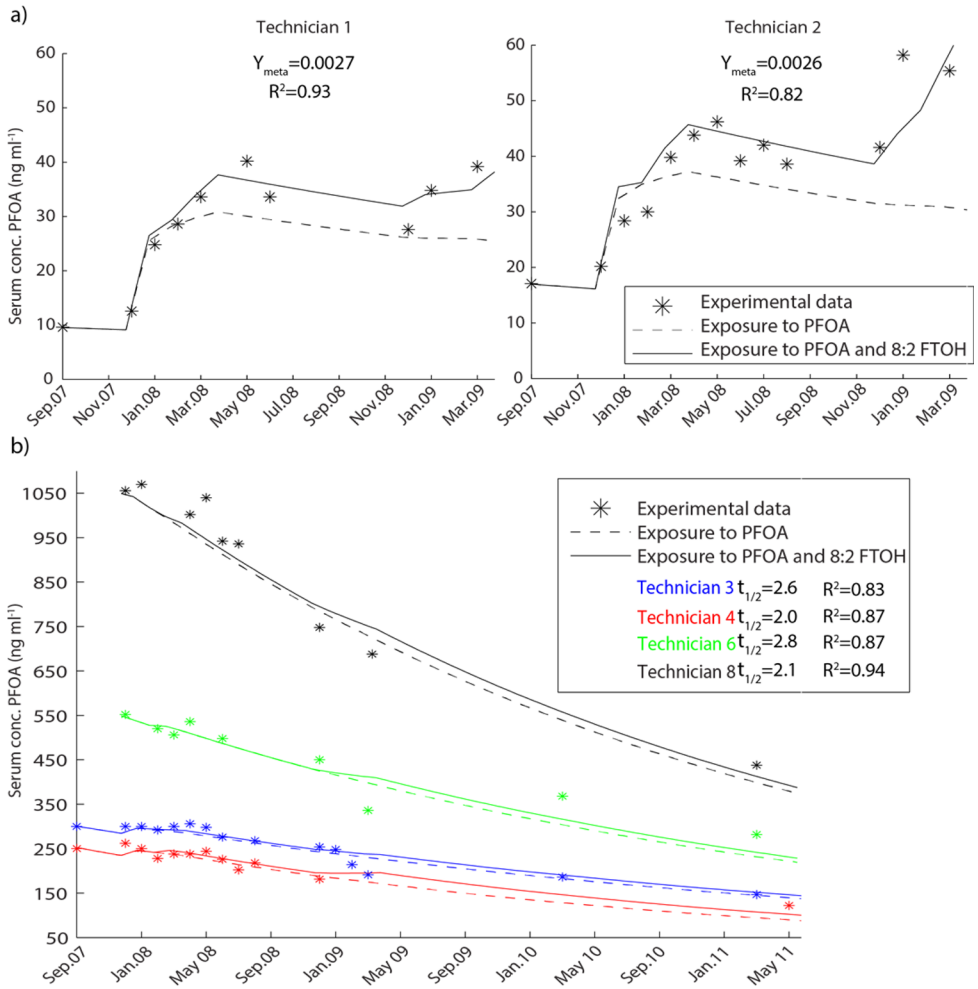

Figure 2. Predicted and measured serum concentrations of PFOA in ng/mL for (a) ski wax technicians with low initial serum concentrations and (b) ski wax technicians with high initial serum concentrations. The dotted and solid lines indicate the model predictions for scenario 1 (exposure to PFOA) and scenario 2 (exposure to PFOA and 8:2 FTOH), respectively. The symbols represent measurements in whole blood by Nilsson et al.[28,29] transformed to serum equivalents. The coefficient of determination (R^2) corresponding to the model including both exposure to PFOA and 8:2 FTOH is indicated for each technician. The fitted metabolism yields (Y_{meta}) and elimination half-lives ($t_{1/2}$, in years) are shown for each technician in (a) and (b), respectively.

The upper and lower limit of the margin of error was obtained by multiplying and dividing, respectively, each fitted Y_{meta} and the $t_{1/2}$ with the corresponding Cf_o obtained with eq 8.

3. RESULTS AND DISCUSSION

The predicted and measured serum concentrations of PFOA for six individual ski wax technicians are displayed in Figure 2. Different trends were observed among the six technicians. For technicians with low initial serum concentrations of PFOA, an accumulation was observed during the ski season followed by slowly decreasing trends in the off-season. The better fit with model scenario 2 demonstrates that both direct and indirect exposure need to be considered to quantitatively predict the absolute concentrations and temporality of PFOA in the serum of technicians with low initial concentrations. For technicians 3, 4, 6, and 8, which had high initial concentrations of PFOA, a net depuration was observed throughout the study period and the occupational exposure during the ski season had a small influence on the PFOA levels. The distinct temporal trends of technicians with low and high initial concentrations of PFOA allowed Y_{meta} and $t_{1/2}$ to be calibrated using different parts of the data set.

3.1. Metabolism of 8:2 FTOH as a Source of PFOA.

Y_{meta} was estimated for the technicians with low initial concentrations only, since their PFOA serum levels were affected by the ongoing exposure. As shown in Figure 2a, the Y_{meta} values obtained from technicians 1 and 2 were similar, 0.0026 to 0.0027, respectively. Because the metabolism yield obtained from *in vitro/in vivo* studies is sometimes expressed as a molar fraction,[26] the corrected Y_{meta} for molar concentrations was 0.0029 and 0.0030 (mean 0.0030). These estimated values are up to 12-fold higher than the metabolism yields obtained from *in vitro* assays in human hepatocytes and microsomes, but in line with those previously reported from *in vivo* studies in rats (0.0025 and 0.014).[23,25,26] The reasons for differences in Y_{meta} reported from various studies have several explanations. In the current study, the variability and uncertainty in external exposure during ski waxing will be propagated to the calculation of Y_{meta}. Accurate determination of concentrations in exposure media together with representative sampling is therefore crucial for determining Y_{meta} in humans (examined further in Section 3.5). In addition, excluding specific exposure routes, such as the hand-to-mouth and dermal exposure pathway, could have altered the final prediction of Y_{meta}. At the same time, the higher

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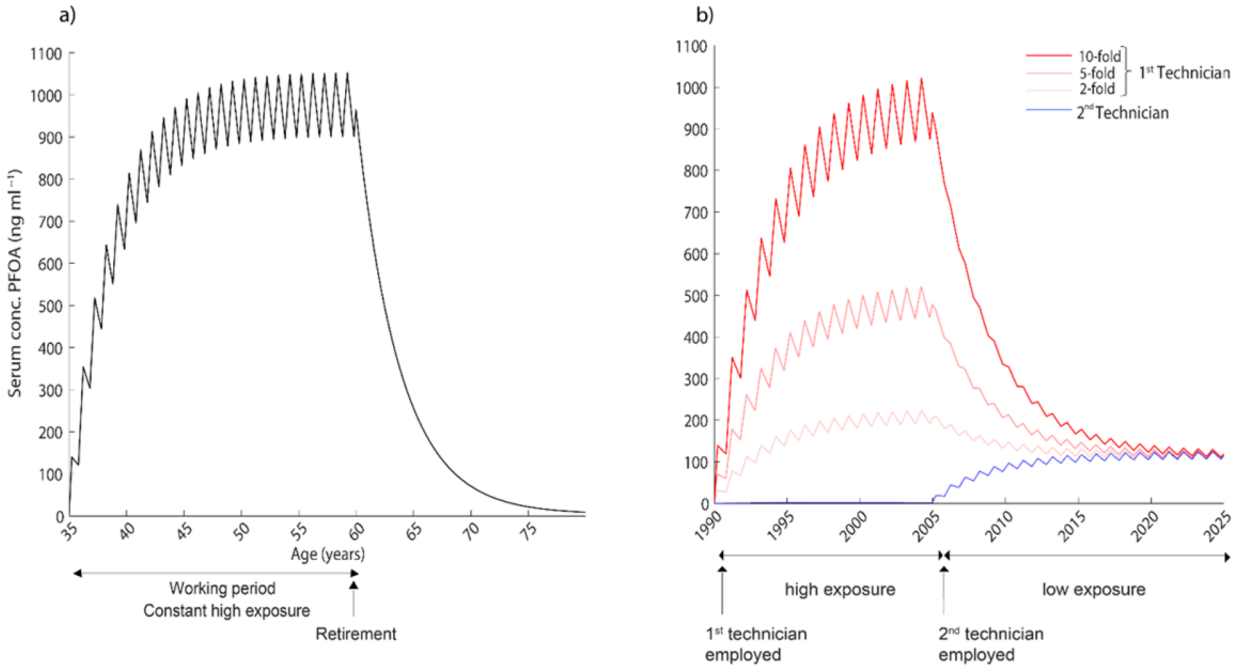

Figure 3. Temporal changes of PFOA in the serum of ski wax technicians illustrated by (a) a technician working for 25 years receiving constant occupational exposure during ski seasons followed by 15 years of retirement (background exposure only) and (b) two hypothetical technicians starting work at different time points (1990 and 2005, respectively) and under different exposure scenarios.

Y_{meta} *in vivo* compared to *in vitro* studies could indicate the existence of additional biotransformation centers than the liver in the human body. The higher doses used in classical biotransformation experiments may also lead to saturation of enzymes involved in the metabolism of FTOHs. Considering all these factors, the Y_{meta} values estimated in this study are probably the most relevant metabolism yields for assessing a real exposure situation for humans.

Even though the averaged Y_{meta} is relatively low, the contribution of indirect exposure was shown to have a clear effect on the PFOA serum concentration of the two ski wax technicians due to high 8:2 FTOH levels in the cabin air. While the model simulations considering exposure to PFOA predicted a stable serum concentration from January 2008 onward, the model simulation considering both PFOA and 8:2 FTOH exposure follows the measured PFOA levels for the two technicians. The exposure to PFOA only (i.e., direct and background exposures) is responsible for at least 55% of the total PFOA concentration in serum and the indirect exposure contributes up to 45% of the PFOA. Furthermore, the PFOA levels resulting from direct exposure is mainly influenced by the amount of PFOA inhaled (i.e., in the air phase) rather than ingested (i.e., sorbed to the suspended particles), which contributes 3% only. Therefore, for the two technicians with low initial concentration, inclusion of indirect exposure is necessary to reproduce the absolute concentrations and temporal trends of PFOA in serum levels over time.

The presence of 7:3 FTCA,[29] another metabolite of 8:2 FTOH, at similar levels (1.6−3.6 ng/mL) in all technicians gives additional evidence that exposure to and metabolism of 8:2 FTOH occurs to a similar extent for all technicians. Therefore, even though the high initial concentrations of PFOA mask the impact of ongoing indirect exposure, the technicians with high initial concentration also appear to metabolize 8:2 FTOH into PFOA, probably with a similar Y_{meta} as the technicians with low initial concentrations.

3.2. Estimated Elimination Half-Life of PFOA. It was possible to estimate the elimination half-life of PFOA from the experimental data of technicians with high initial concentration since they depurated PFOA throughout the monitoring period and ongoing direct/indirect occupational exposure had no major impact on PFOA serum levels. For technician 8, however, the serum concentration data point from September 2007 was not included. It appears to be an outlier since fitting was poor when this data point was included (see Appendix 5 of the Supporting Information). The elimination half-lives estimated in this study are "intrinsic" (i.e., corrected for the ongoing exposure), as opposed to "apparent".[33] As shown in Figure 2b, the fitted elimination half-lives ranged between 2.0 and 2.8 years, and the average (2.4 years) is the same as the averaged intrinsic elimination half-life of 2.4 years estimated by Russell et al.[33] Similar elimination half-lives were also reported by Bartell et al. (2.2 to 2.6 years)[58] and Olsen et al. (3.5 years) although interindividual variability ranging from 1.5 to 9.7 years has been observed among occupationally exposed men.[52] Compared to rats and mice, which have PFOA half-lives of hours to weeks, humans have a slower elimination rate with an elimination half-life counted in years.[61] The slow elimination in humans is thought to be caused by a strong affinity to serum albumin,[9] enterohepatic circulation,[62] and reabsorption of PFOA from kidney filtrate by organic anion transporters.[11] Given that the elimination of PFOA is governed by several protein-specific interactions, a pharmacokinetic model including a saturable resorption mechanism have been proposed.[31,32] However, according to the calibration in Figure 2b and other studies on highly exposed humans,[63,64] the elimination rate of PFOA appears to be independent of serum concentrations, indicating that saturation is unlikely to happen under realistic exposure conditions. In addition to urinary and fecal excretion, PFOA is believed to be eliminated through menstruation, resulting in a lower elimination half-life of PFOA in women compared to men.[51] Since all technicians in this study were

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men, this elimination mechanism was not considered in the model.

3.3. Interpretation of Biomonitoring Time Trends in Ski Wax Technicians. For technicians with low initial concentrations (Figure 2a), PFOA body burdens increased during the working seasons and decreased during the unexposed periods. While the model predictions of PFOA serum concentrations were in good agreement with the measured data until March 2009 (R^2 varies from 0.82 to 0.93), it was not possible to run the model for a longer period due to a lack of information on the external exposure. The reasonable fit of the model to the biomonitoring data suggests that PFOA and 8:2 FTOH concentrations monitored by Nilsson et al.[28,29] in the cabin are representative of the serum concentrations of the two technicians.

For the four technicians with high initial serum concentrations (Figure 2b), the model predictions were fairly close to the measured concentrations, with R^2 between 0.83 and 0.94. As opposed to the group with low initial concentrations, occupational exposure (both direct and indirect) had a minor influence on the PFOA serum concentration trends. The different external exposures experienced by each technician (see Appendix 4 of the Supporting Information) were not able to explain the differences in serum concentrations in both technician groups. Consequently, despite the lack of data on external exposure between 2010 and 2011,[29] the model was successfully run over the 4-year period. These technicians likely experienced a higher exposure in the years prior to the commencement of monitoring by Nilsson et al.[28] Furthermore, the concentrations of PFOA in serum were shown to be proportional to the amount of years that the technicians had worked (see Table A2-1 in the Supporting Information for the work history of each technician). The relative influence of ongoing exposure on the serum time trends depended therefore on the duration and magnitude of historical exposure. At the end of the study (May 2011), the technicians with serum concentrations close to the new steady-state (technicians 3 and 4) had a slower apparent elimination rate compared to technician 8, since the external exposure was relatively more influential.

3.4. Temporal Trends of PFOA Serum Levels in Evaluative Exposure Scenarios. The evaluative model simulations illustrate the long-term accumulation/depuration of PFOA serum levels expected in ski wax technicians under different exposure scenarios (Figure 3) and throughout their working life. The seasonal exposure is reflected by the serum concentration of PFOA, which increases stepwise every winter and decreases in the off-season, as observed in Figure 2a for technicians with low initial concentration.

In Figure 3a, the seasonal exposure to PFOA and 8:2 FTOH results in an increase of the serum concentration of PFOA that is faster during the first years of work and then slows down as steady-state is reached after about 15 years of exposure. Once the technician retires, background exposure, which is 3 orders of magnitude lower than the occupational exposure, becomes the only exposure source of PFOA. As a consequence, depuration of PFOA takes place until reaching a new steady-state serum concentration with the background external exposure at approximately 2 ng/mL. This demonstrates that the occupational exposure of ski waxers is much more important than the background exposure in defining the serum levels of PFOA over time. Consequently, even though the background exposure is, in reality, not constant, accounting for this variability would have a very small effect on the model results.

Figure 3b shows the consequence of a change in the external exposure on the serum levels of two ski wax technicians with different exposure scenarios. The first technician worked for 15 years under high exposure allowing the concentration of PFOA in serum to approach steady-state, in all three different exposure conditions. When the second technician starts to work, the intake levels are up to ten times lower than the previous exposure conditions, reflecting an introduction of exposure mitigation measures (e.g., protective equipment and increased ventilation). This sudden shift in external exposure results in a readjustment of the serum concentration through depuration of PFOA for the first technician and an increase of the serum PFOA of the second technician, who was at steady-state with the background exposure intake during the pre-employment period. This simulation is representative of the temporal trends observed for the six ski wax technicians in Nilsson et al.'s study.[28,29] In addition, it corroborates the hypothesis presented in Section 3.3 that the PFOA levels in serum of the ski wax technicians and their time trends are determined by (1) the duration of exposure (i.e., number of years working as a technician), as suggested by Nilsson et al.[28] and Freberg et al.[20] and (2) the different external exposure levels to which the technicians are exposed over time. The lessons learned from these evaluative scenarios for hypothetical technicians are valuable for understanding the occupational exposure of the real ski wax technicians.

Finally, comparing Figure 3a,b, the apparent elimination is slower in the second simulation since the readjusted occupational exposure is much higher than the background exposure in the first simulation. Consequently, as highlighted by Russell et al.,[33] calculating the "intrinsic" elimination half-life from the depuration phase without considering the ongoing albeit lower exposure will result in an erroneously long "apparent" elimination half-life.

3.5. Sensitivity and Uncertainty Analysis. The sensitivity and uncertainty analysis of the fitted Y_{meta} and $t_{1/2}$ were performed on each technician (e.g., technicians with low initial concentrations for Y_{meta} and technicians with high concentrations for $t_{1/2}$). For technicians with low initial concentrations, the input parameters defining the intake (inhalation rate, uptake efficiency, working hours), the concentrations of PFOA in air, and the input parameters defining the amount of chemical partitioning in serum (volume of distribution, body weight) were the most important parameters for determining Y_{meta}, further indicating the importance of ongoing exposure for this occupationally exposed group. The exposure to dust was, however, a minor contributor for the fitted output parameter. For the other technician group, the initial concentration of PFOA in serum becomes dominant in defining the elimination half-life of PFOA, as also illustrated by the additional simulation of technician 8 (see Appendix 5 in the Supporting Information). The results from the sensitivity analysis support the conclusions made in previous sections: the corresponding fitted output parameter is mainly determined by the initial levels of PFOA in serum (i.e., the past exposure conditions) for technicians with high initial concentration and by the ongoing exposure for the technicians with low initial concentration.

The results of the uncertainty analysis are presented in Figure A6-2 and Table A6-2 of the Supporting Information. The concentration of PFOA in air and the assumed inhalation rate were mainly responsible for the uncertainty of Y_{meta}, and

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the initial concentrations and the volume of distribution were mainly contributing to the uncertainty of $t_{1/2}$. An error range of 0.0006 to 0.01 (molar concentration basis) was estimated for Y_{meta} and 1.8 to 3.1 years for $t_{1/2}$. The confidence factors are higher for Y_{meta} (i.e., average of 5.2) than for $t_{1/2}$ (i.e., average of 1.3) indicating that the Y_{meta} values estimated in this study have a larger uncertainty than the estimated $t_{1/2}$.

Although the number of individuals on which Y_{meta} was calibrated was low (n = 2), the substantial amount of data available for each individual covered a sufficiently long time period to characterize the temporal trends of PFOA in serum. The quantification of exposure, which will have a direct impact on the calibrated Y_{meta}, can also vary depending on the sampling method used (i.e., personal pump versus platform pump). In this study, measurements obtained from personal pumps were used since they reflect the breathing zone of occupational ski waxers and this should provide a good estimate of total exposure for these highly exposed individuals. Following these arguments and the fact that the model-derived metabolism yields compared well to those previously obtained from *in vivo* and *in vitro* studies,[23,25,26] the estimated range of values of Y_{meta} (mean 0.003, molar concentration basis; uncertainty range of 0.0006−0.01) presented in this paper appear reasonable.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.est.6b01477.

Additional information on the ski wax technicians and Nilsson et al.'s study,[28,29] the treatment and variability in external exposure measurements, the additional model run for technician 8, and the results related to the sensitivity and error analysis are available in the Supporting Information (PDF)

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Notes

The authors declare no competing financial interest.

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